

Road Accident Severity & Safety Hotspot Analyzer

Predicting accident severity with machine learning, uncovering high-risk patterns, and turning 12,316 real accident records into actionable road-safety recommendations.

12,316
RECORDS

32
FEATURES

3
ML MODELS

84.58%
BEST ACCURACY

Team **Komal Kumari** · Pallavi · Shruti Saini

Dataset: Kaggle — Road Accidents Severity (RTA) | github.com/KomalKumari27

OVERVIEW

Agenda

How this presentation is structured

01

Problem & Objectives

Why accident severity prediction matters

02

Dataset at a Glance

12,316 records · 32 features · 3 severity classes

03

Tech Stack & Workflow

Tools, project structure and the 8-step pipeline

04

Exploratory Data Analysis

When, who, where and why accidents happen

05

Model Training & Evaluation

Logistic Regression vs Decision Tree vs Random Forest

06

Feature Importance

What actually drives severity

07

Hotspots & Recommendations

High-risk patterns → concrete safety actions

08

Insights, Limitations & Future Scope

What we learned and where this goes next

THE CHALLENGE

Problem Statement & Objectives

THE PROBLEM

Road accidents are among the leading causes of injuries and fatalities worldwide. Authorities react after crashes happen — they lack tools to anticipate which conditions turn an ordinary trip into a severe accident.

Understanding the factors behind accident severity lets governments and transport authorities act preventively — before lives are lost.



Predict severity

Classify each accident as Slight, Serious or Fatal from conditions known at the scene.



Identify high-risk patterns

Mine the data for combinations of weather, light, road and driver factors that raise severity.



Explain the drivers

Rank the most influential accident factors using model feature importance.



Recommend actions

Translate findings into concrete safety recommendations for decision-makers.

WHY IT MATTERS

Real-World Impact & Stakeholders

Data-driven road-safety planning for the people who need it



Traffic Police

Prioritise patrols and enforcement where severe accidents are most likely.



Road Safety Authorities

Target lighting, signage and surface maintenance to high-risk segments.



Government Agencies

Ground policy and budget decisions in evidence, not anecdotes.



Emergency Response

Anticipate severe-accident conditions to position responders faster.



Transport Departments

Design awareness campaigns for the riskiest driver groups and behaviours.

The goal: shift road safety from reactive investigation to proactive prevention using patterns hidden in historical accident data.

THE DATA

Dataset at a Glance

Road Traffic Accident (RTA) dataset — Kaggle

12,316

ACCIDENT RECORDS

32

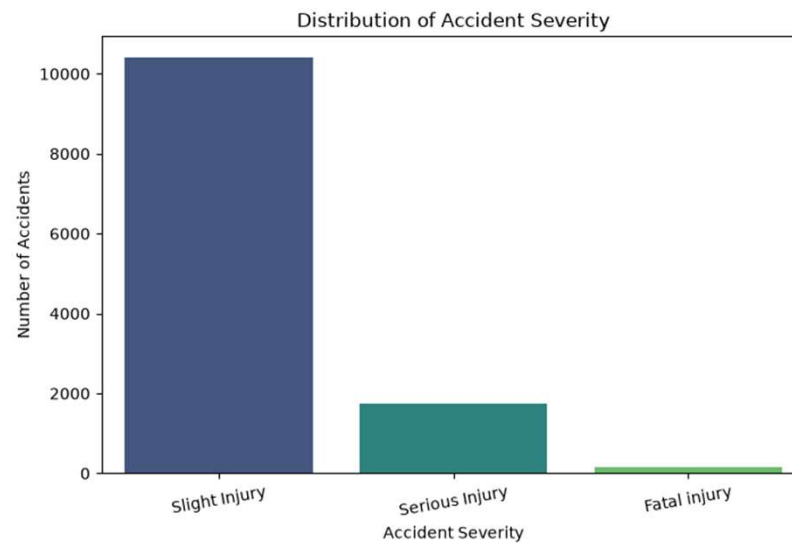
RAW FEATURES

3

SEVERITY CLASSES

80 / 20

SPLIT — TRAIN 9,852 · TEST 2,464



Target variable — Distribution of Accident Severity

CLASS BALANCE

■ Slight Injury
≈ 84.5%

■ Serious Injury
≈ 14.1%

■ Fatal Injury
≈ 1.3%

Heavily imbalanced — a key challenge for modelling and evaluation.

Features cover driver (age band, experience), vehicle, environment (weather, light, road surface) and crash context (cause, vehicles involved, time, day).

FOUNDATIONS

Tech Stack & Project Structure

TECHNOLOGIES USED

Python

Pandas

NumPy

Scikit-learn

Matplotlib

Seaborn

Jupyter Notebook

Git & GitHub

WHAT EACH TOOL DOES

- **Pandas / NumPy** — cleaning, encoding, feature engineering
- **Seaborn / Matplotlib** — 9+ EDA visualisations & severity cross-tabs
- **Scikit-learn** — model training, metrics, feature importance
- **Jupyter** — 3 staged notebooks: EDA → Training → Hotspots

REPOSITORY LAYOUT

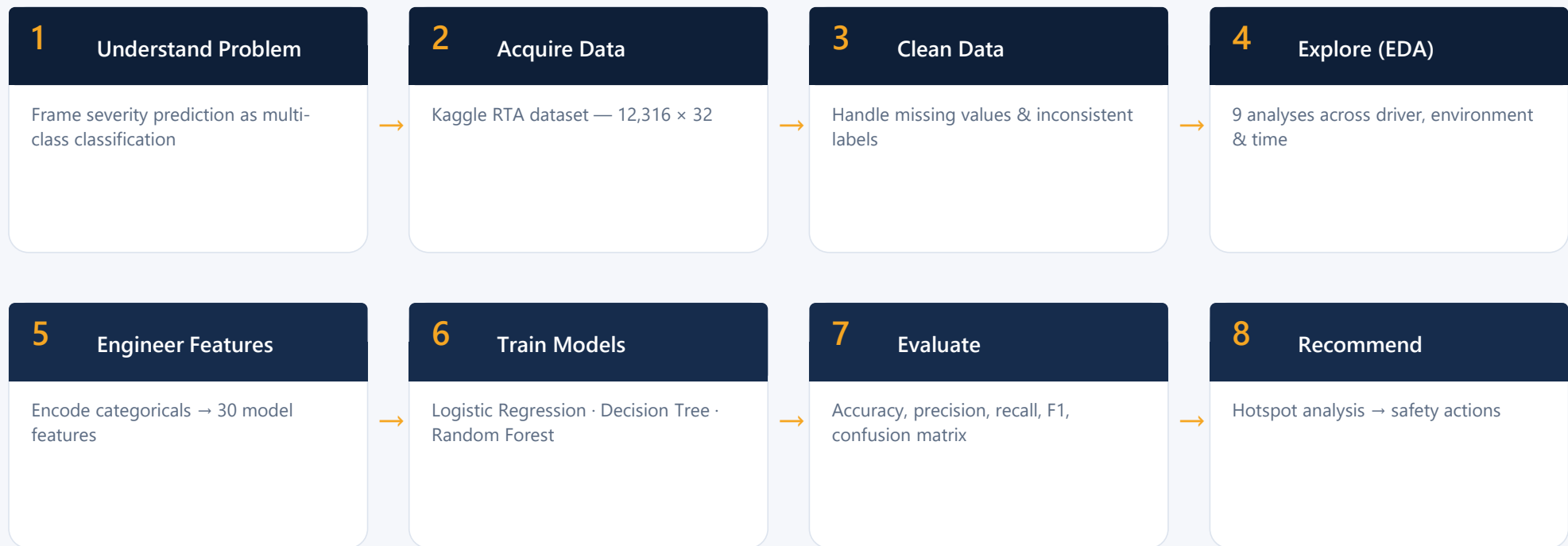
road_accident_severity_analyzer/

```
├─ data/                RTA Dataset.csv · X_train · X_test
├─ notebook/
│   ├── 01_eda.ipynb      exploratory analysis
│   ├── 02_Model_Training.ipynb  3 classifiers + metrics
│   └─ 03_Hotspot_Analysis.ipynb  patterns & actions
├─ output/
│   ├── safety_recommendations.csv
│   └─ screenshots/      13 result charts
├─ docs/                project report (PDF)
├─ requirements.txt
└─ README.md
```

METHODOLOGY

End-to-End Project Workflow

Eight stages from raw CSV to safety recommendations



Each stage lives in its own notebook, so the analysis is reproducible step by step.

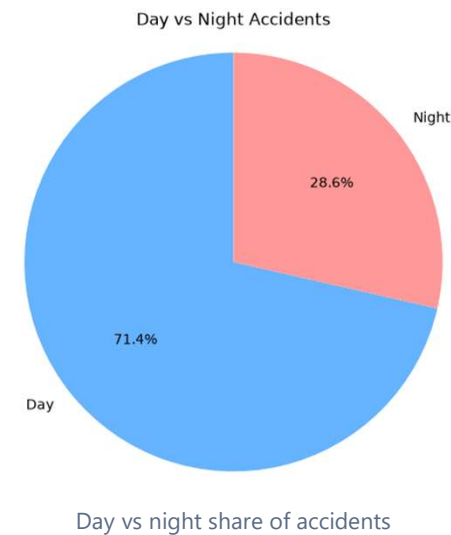
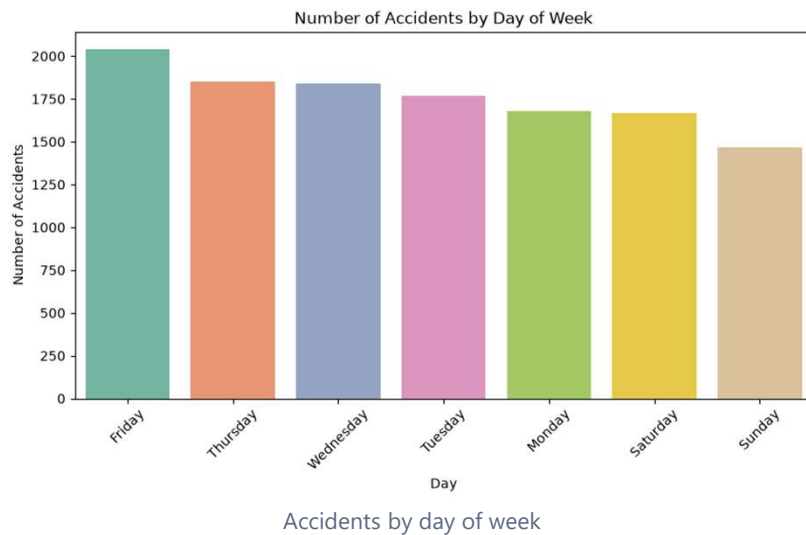
01

Exploratory Data Analysis

What 12,316 accidents reveal about when, who and why

- Severity & time patterns
- Driver age & experience
- Weather, light & road surface
- Top causes of accidents

When Do Accidents Happen?



STANDOUTS

Friday

peak day (~2,040)

Sunday

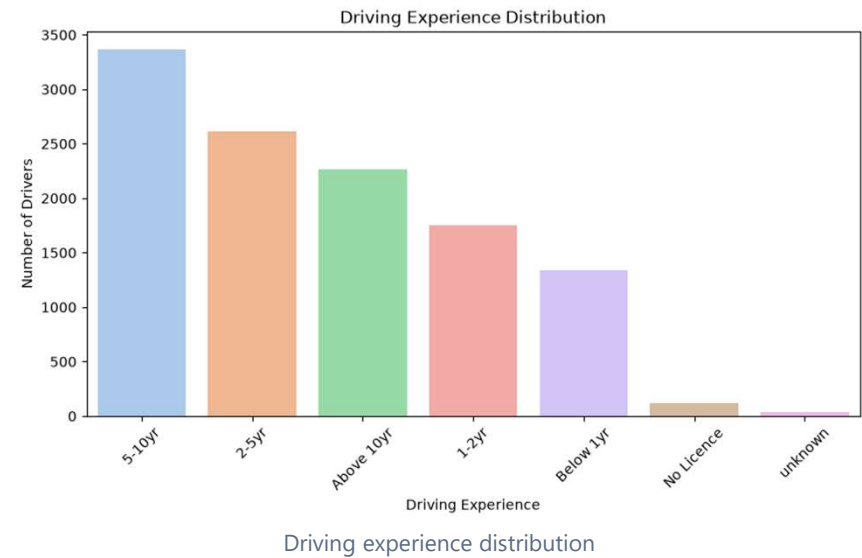
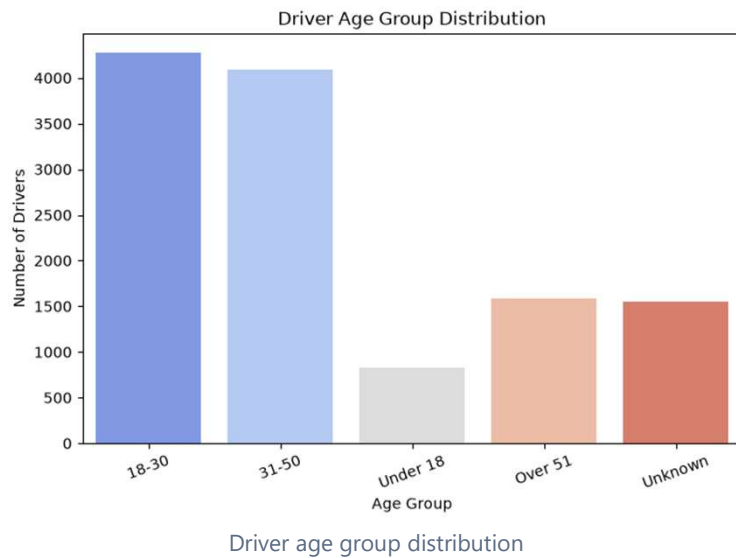
quietest day

71.4%

occur in daylight

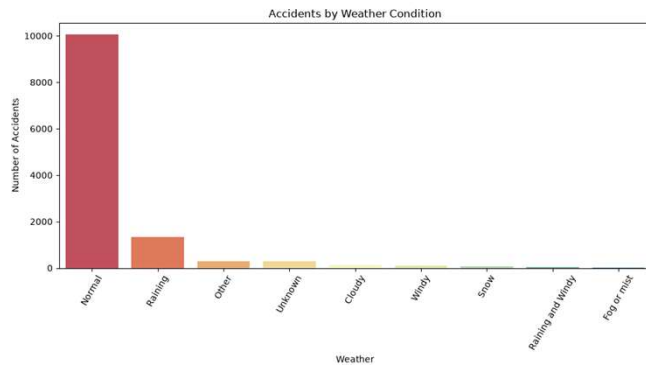
- **Friday leads:** end-of-week commuting and higher traffic volume drive the weekly peak; weekdays consistently beat weekends.
- **Daylight dominates counts,** but night driving carries higher severity risk per trip — reduced visibility and fatigue — which motivates the lighting analysis later.

Who Is Involved?

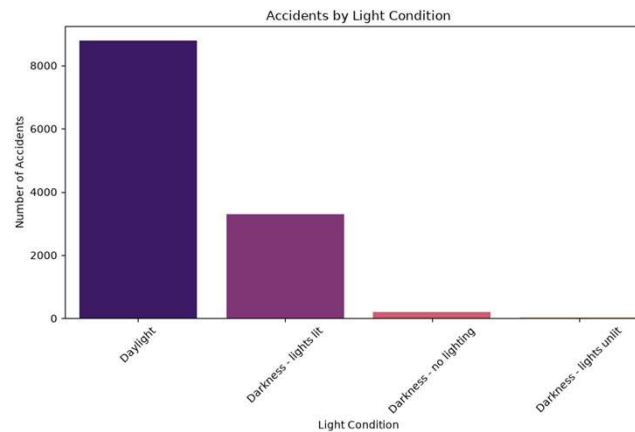


- **Ages 18–50 dominate** (two largest bands ~4,200 each) — the most active driving population is also the most exposed.
- **Experience doesn't immunise:** drivers with 5–10 years of experience top the chart (~3,350), showing over-confidence matters as much as inexperience.

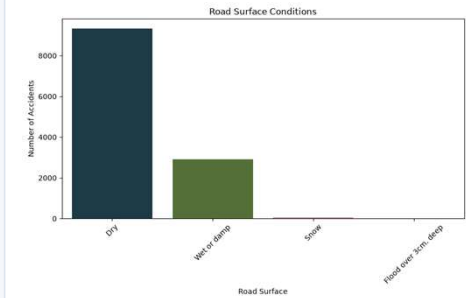
Road, Weather & Light Conditions



Accidents by weather



Accidents by light condition

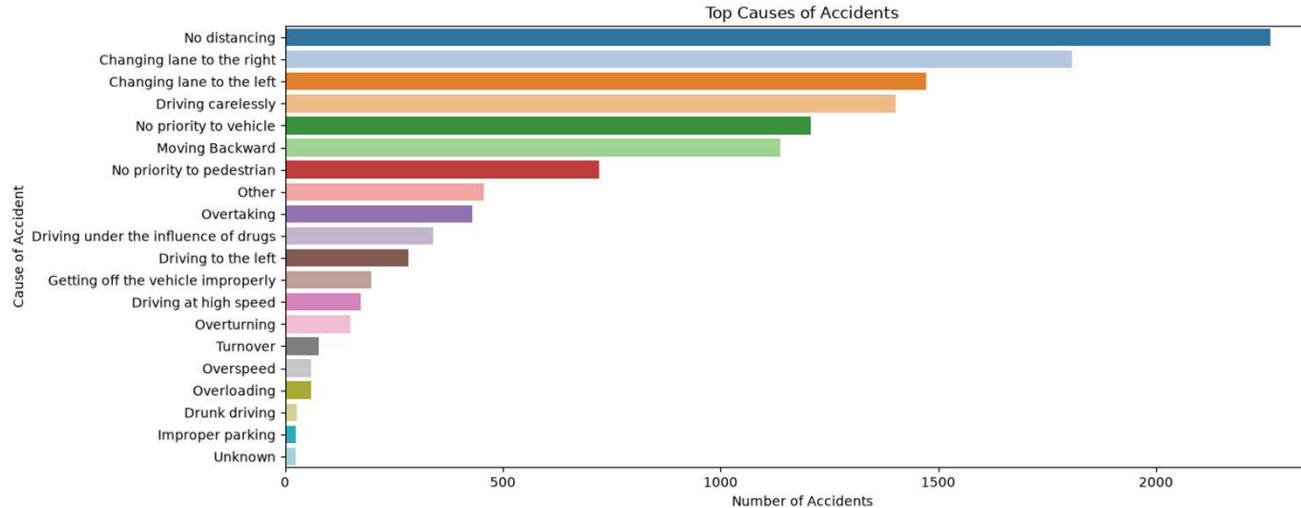


Road surface conditions

- Most accidents happen in 'normal' conditions (clear weather, dry roads, daylight) — simply because that's when most driving happens.
- Rain and wet/damp surfaces are the leading adverse conditions and show a higher share of serious outcomes — skidding, longer braking distances.
- Darkness with lights lit still accounts for ~3,300 accidents — poor lighting is a lever authorities directly control.

Why Do Accidents Happen?

Top causes of accidents — driver behaviour dominates



Cause of accident, ranked by frequency

TOP 5 CAUSES

- 1 No distancing ~2,260
- 2 Lane change (right) ~1,810
- 3 Lane change (left) ~1,470
- 4 Careless driving ~1,400
- 5 No priority to vehicle ~1,210

Nearly all top causes are behavioural → preventable.

Tailgating and unsafe lane changes alone account for thousands of accidents — awareness campaigns and enforcement can attack these directly.

02

Modelling & Evaluation

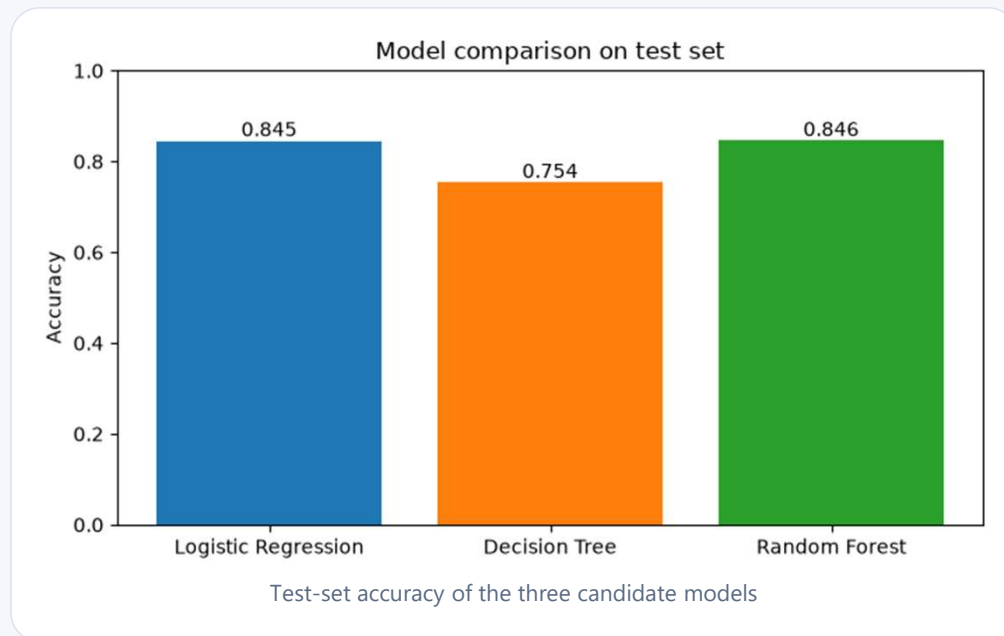
Three classifiers, one winner — and an honest look at its errors

- Feature engineering & split
- Model comparison
- Accuracy · Precision · Recall · F1
- Confusion matrix & feature importance

MODELLING

Training & Model Comparison

9,852 training rows · 30 engineered features · evaluated on 2,464 held-out rows



Logistic Regression

84.54%

Strong linear baseline — nearly matches the winner

Decision Tree

75.37%

Overfits the training data; weakest generalisation

Random Forest

84.58%

Best test accuracy + robustness + feature importance

✓ SELECTED MODEL

Random Forest was selected: it edges out Logistic Regression on accuracy while providing ensemble robustness and interpretable feature importances.

EVALUATION

How Good Is the Model — Really?

Random Forest on the 2,464-row test set (weighted metrics)

84.58%

ACCURACY

0.7795

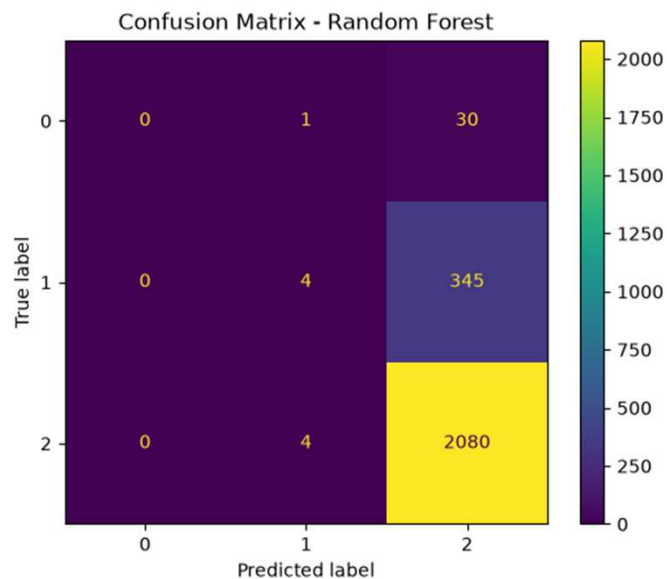
PRECISION (WEIGHTED)

0.8458

RECALL (WEIGHTED)

0.7783

F1 SCORE (WEIGHTED)



Confusion matrix (0=Fatal · 1=Serious · 2=Slight)

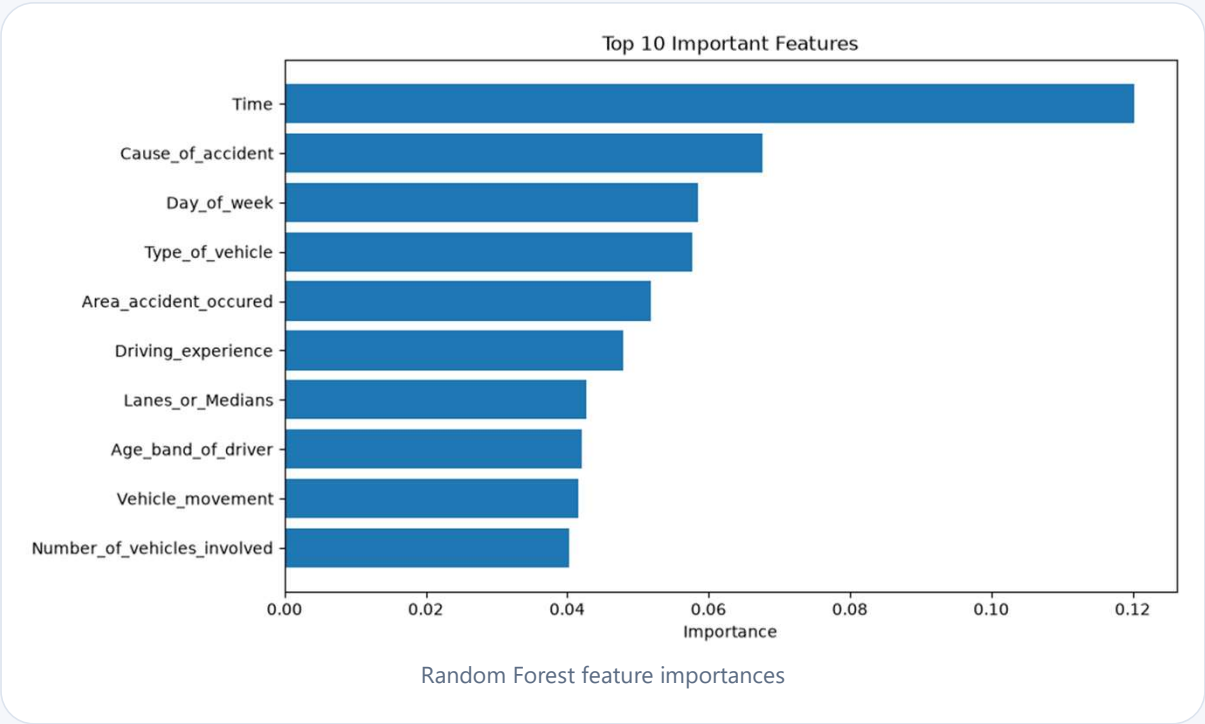
HONEST READING

- **Slight injuries:** 2,080 / 2,088 correctly caught — excellent.
- **Minority classes suffer:** most Serious and Fatal cases are pulled toward the majority class.
- **Root cause:** class imbalance (≈85% slight) — accuracy alone overstates performance.
- **Next step:** SMOTE / class weights to lift minority recall.

EXPLAINABILITY

What Drives Accident Severity?

Top 10 features by Random Forest importance





Time of day

By far the strongest signal (0.120) — severity risk shifts across the day.



Cause of accident

Behavioural causes (0.068) directly shape outcomes.



Day of week & vehicle type

Traffic rhythm and vehicle class both matter (~0.058 each).



Driver factors

Experience and age band appear in the top 10 — confirming the EDA.

Environment sets the stage, but time and human behaviour dominate — exactly where policy interventions are cheapest.

03

Hotspots & Recommendations

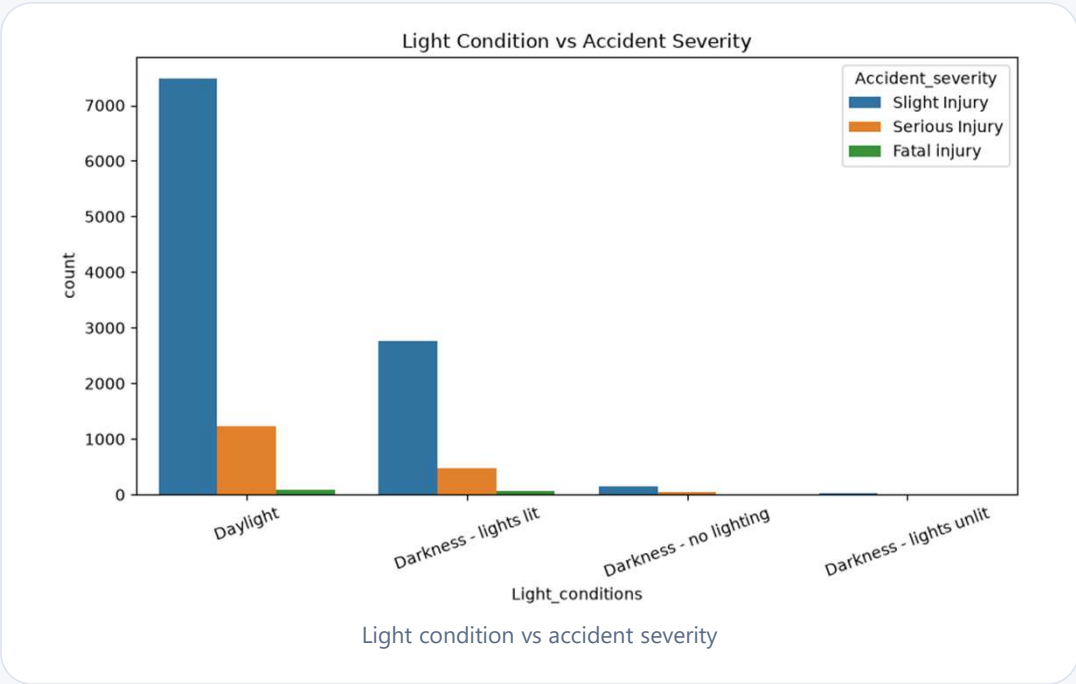
From patterns in the data to actions on the road

- Severity vs light conditions
- High-risk pattern summary
- Concrete safety actions
- Limitations & future scope

HOTSPOT ANALYSIS

Where Severity Concentrates

Cross-tabulating conditions against severity reveals high-risk scenarios



Poor lighting

Darkness — even with lights lit — carries a higher share of serious & fatal outcomes.

Adverse weather

Rain shows a disproportionately serious accident profile despite fewer total crashes.

Wet / damp surfaces

Different accident pattern vs dry roads: skidding & reduced grip raise severity.

Risky driver groups

Young drivers and behavioural causes (tailgating, lane-changes) recur across severe cases.

A 'hotspot' here is a condition pattern, not a GPS location — the dataset has no coordinates, so the analysis targets when and how, not where on the map.

DELIVERABLE

Safety Recommendations

Generated by the analysis and exported to safety_recommendations.csv

	RISK FACTOR	RECOMMENDED ACTION	PRIMARY OWNER
	Rainy weather	Reduce speed limits during rain	Traffic authorities
	Poor lighting	Install additional street lights	Road safety dept.
	Driver inattention	Increase awareness campaigns	Government / media
	Wet roads	Improve road maintenance	Public works
	Young drivers	Provide defensive driving training	Licensing bodies

Each recommendation maps a data-confirmed risk factor to a low-cost, high-leverage intervention that authorities can act on immediately.

Key Insights & Honest Limitations

✓ KEY INSIGHTS

- **85% of accidents cause slight injuries** — but the rare severe ones are what matter most.
- **Human behaviour is the biggest lever:** tailgating and lane changes top the cause list.
- **Time of day is the #1 predictive feature** for severity.
- **Random Forest wins** at 84.58% test accuracy over LR (84.54%) and DT (75.37%).
- **Mid-experience drivers (5–10 yrs)** crash the most — experience ≠ safety.

⚠ LIMITATIONS

- **Historical data only** — no real-time traffic or weather feeds.
- **No GPS coordinates**, so hotspots are condition patterns, not map locations.
- **Class imbalance** depresses minority-class (Fatal/Serious) precision and F1.
- **Data quality dependent:** 'Unknown' categories add noise.
- **Decision support, not replacement** for expert judgement.

ROADMAP

Future Scope

Where this project goes next



NEXT UP

Streamlit dashboard

Enter accident details → predict severity → view recommendations, live.



MODELLING

Handle imbalance

SMOTE / class weights + XGBoost & LightGBM to lift Fatal-class recall.



DATA

GPS hotspot mapping

Geo-coded data to visualise true map-based accident hotspots.



DATA

Live API integration

Real-time weather and traffic feeds for on-the-fly risk scoring.



MODELLING

Deep learning

Neural approaches once richer data is available.



DEPLOYMENT

Cloud + mobile

Deployed service and mobile app for field use by authorities.

Thank You

Every severe accident predicted is a chance to prevent one. This project shows how open data and simple, well-evaluated machine learning can guide real road-safety decisions.

TEAM

Komal Kumari

Pallavi

Shruti Saini

PROGRAM

IIT Mandi

Himshikhar Project Program

RESOURCES

Kaggle RTA Dataset

github.com/Komalkumari27

scikit-learn · pandas · seaborn